SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

DRAWING REGISTER AND ENGINEERING STATUS

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

PACKAGE PURPOSE

Develop existing bridge concepts into a traceable preliminary engineering package.

Use confirmed criteria and source drawings without inventing unverified site or fabrication dimensions.

Three arrangement sheets + three bearing/interface sheets + three detailing schedules.

Six original tender landing sheets are appended unchanged after the development sheets.

Open engineering inputs and calculations are recorded in the companion report and workbook.

Proposed ID	Bridge	Sheet scope	Status
SHV-BRRP-101-GA-P4	ICW-ICG LD	Study-source arrangement	PRELIMINARY
SHV-BRRP-102-GA-P4	ICP-ICG LD	Study-source arrangement	PRELIMINARY
SHV-BRRP-103-GA-P4	BHS-SLQ LD	Study-source arrangement	PRELIMINARY
SHV-BRRP-101-BI-P4	ICW-ICG LD	Bearing / landing interface	PRELIMINARY
SHV-BRRP-102-BI-P4	ICP-ICG LD	Bearing / landing interface	PRELIMINARY
SHV-BRRP-103-BI-P4	BHS-SLQ LD	Bearing / landing interface	PRELIMINARY
SHV-BRRP-101-DD-P4	ICW-ICG LD	Detail development schedule	PRELIMINARY
SHV-BRRP-102-DD-P4	ICP-ICG LD	Detail development schedule	PRELIMINARY
SHV-BRRP-103-DD-P4	BHS-SLQ LD	Detail development schedule	PRELIMINARY

COMPUTATION BOUNDARY

Computed: loading arithmetic, source deflection ratios, thermal sensitivity, section geometry and ideal wrap quantities.

Not computed: a new global structural model, code member resistances, connection/bearing capacities or marine-operation design.

The detailed drawing register identifies the calculations and measurements required to close each detail.

SHV-BRRP-P4-INDEX | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

Basis: DC3.4 PartI Rev14 / BRRP PartII / B774 scope, addendum and tender sources

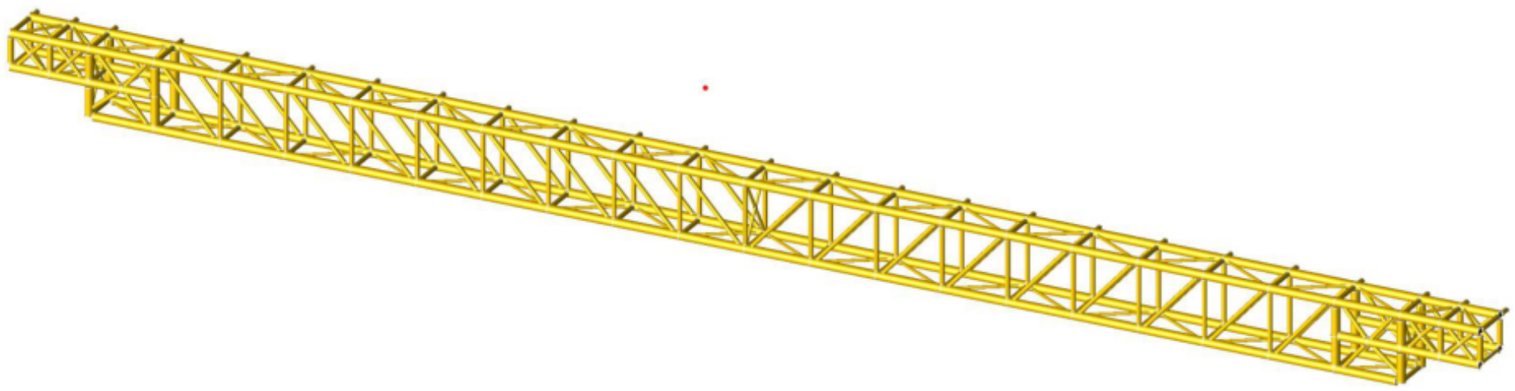
No member resistance, connection, bearing or lift approval is implied. Source results are not a new independently solved analysis.

SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

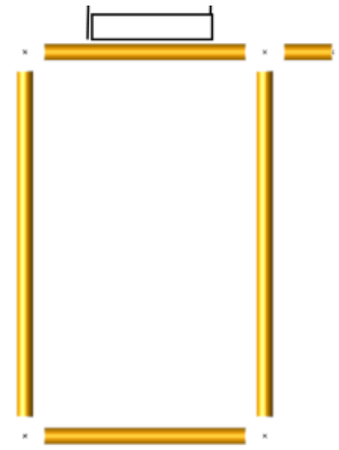
ICW-ICG LD - PRELIMINARY ARRANGEMENT

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

SOURCE-STUDY THREE-DIMENSIONAL MODEL



SOURCE CROSS-SECTION



Structural depth / overall width: not verified

Loading-width reference: 1.2 m

LONGITUDINAL STUDY REFERENCE - NOT TO SCALE

Study model length 56 m; approximate reported gap 54.5 m

ICW - sliding

ICG - hinged/fixed intent

Final bearing centres, levels, skew, member coordinates, section mapping and camber profile require engineering verification.

25-year design life. E=210000 MPa; density=7850 kg/m3; alpha=12x10^-6 /C. (Criteria pp66,119)

Global walkway loading2.4525kN/m2; local grating loading4.905kN/m2. Actual loading areas must be reconciled.

Study source vertical deflection 105mm / criteria limit 140mm; lateral 87mm / 112mm.

Wind gust duration: 15sec per Annexure5. Speed and elevation correction require documented calculation.

Camber to compensate verified dead-load deflection; source1:200 statement is not adopted.

Include approved landing modifications and applicable ICG/ICP jacket strengthening.

SHV-BRRP-101-GA-P4 | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

Source B774-000-83-43-BPR-0004 Rev0; combined SOW/report PDF pp.133-135;152-153

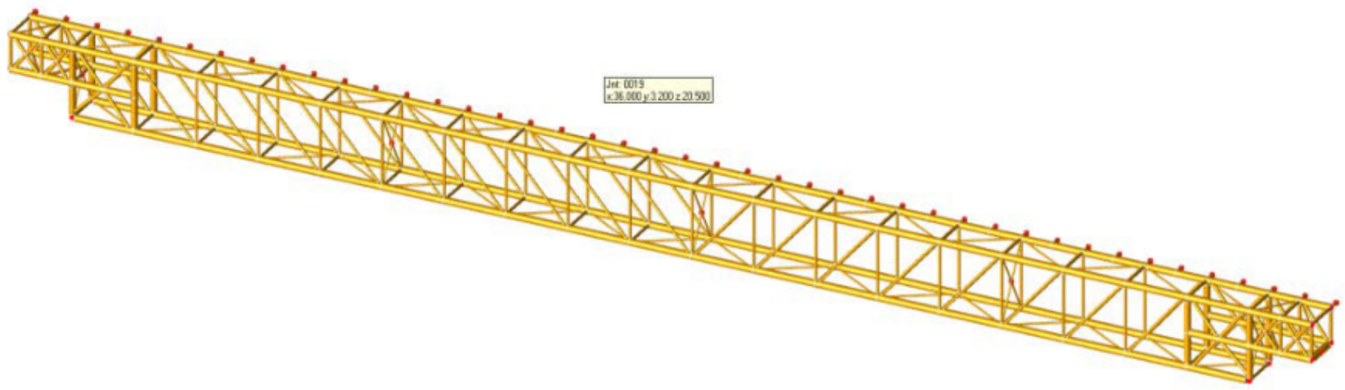
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SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

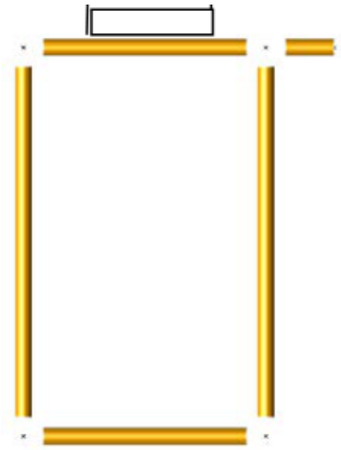
ICP-ICG LD - PRELIMINARY ARRANGEMENT

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

SOURCE-STUDY THREE-DIMENSIONAL MODEL



SOURCE CROSS-SECTION



Structural depth / overall width: not verified

Loading-width reference: 1.2 m

LONGITUDINAL STUDY REFERENCE - NOT TO SCALE

Study model length 84 m; approximate reported gap 80.5 m

ICP - sliding

ICG - hinged/fixed intent

Final bearing centres, levels, skew, member coordinates, section mapping and camber profile require engineering verification.

25-year design life. E=210000 MPa; density=7850 kg/m3; alpha=12x10^-6 /C. (Criteria pp66,119)

Global walkway loading2.4525kN/m2; local grating loading4.905kN/m2. Actual loading areas must be reconciled.

Study source vertical deflection 177mm / criteria limit 210mm; lateral 123mm / 168mm.

Wind gust duration: 15sec per Annexure5. Speed and elevation correction require documented calculation.

Camber to compensate verified dead-load deflection; source1:200 statement is not adopted.

Include approved landing modifications and applicable ICG/ICP jacket strengthening.

SHV-BRRP-102-GA-P4 | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

Source B774-000-83-43-BPR-0003 Rev0; combined SOW/report PDF pp.196-199;207-209

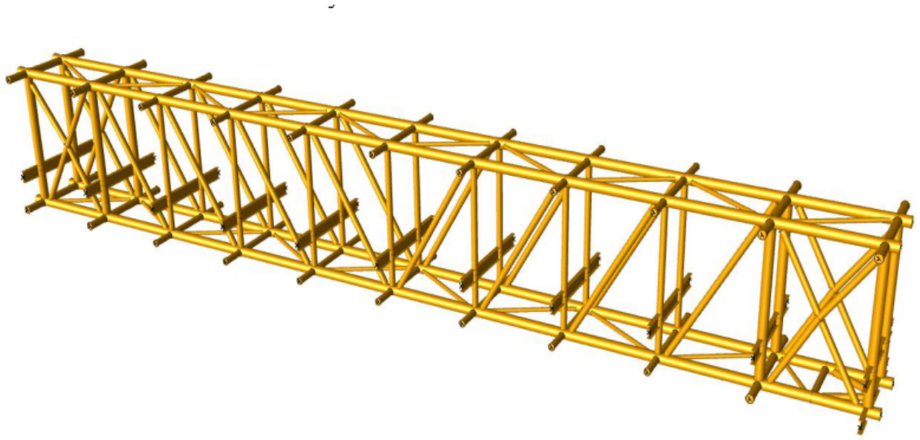
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SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

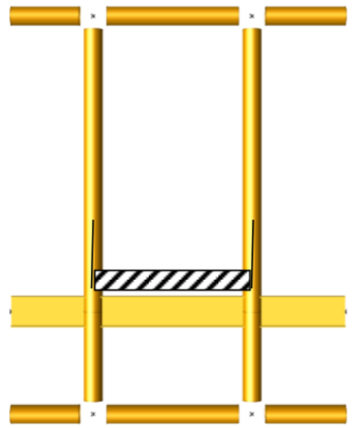
BHS-SLQ LD - PRELIMINARY ARRANGEMENT

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

SOURCE-STUDY THREE-DIMENSIONAL MODEL



SOURCE CROSS-SECTION



Structural depth / overall width: not verified

Loading-width reference: 2.324 m

LONGITUDINAL STUDY REFERENCE - NOT TO SCALE

Study model length 42 m; approximate reported gap 40 m

BHS - sliding

SLQ - hinged/fixed intent

Final bearing centres, levels, skew, member coordinates, section mapping and camber profile require engineering verification.

25-year design life. E=210000 MPa; density=7850 kg/m3; alpha=12x10^-6 /C. (Criteria pp66,119)

Global walkway loading2.4525kN/m2; local grating loading4.905kN/m2. Actual loading areas must be reconciled.

Study source vertical deflection 32mm / criteria limit 105mm; lateral 76mm / 84mm.

Wind gust duration: 3sec per Annexure5. Speed and elevation correction require documented calculation.

Camber to compensate verified dead-load deflection; source1:200 statement is not adopted.

BHS new bridge and services must be accepted before existing bridge demolition.

SHV-BRRP-103-GA-P4 | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

Source B774-000-83-43-BPR-0005 RevA; combined SOW/report PDF pp.97-100;122-123

No member resistance, connection, bearing or lift approval is implied. Source results are not a new independently solved analysis.

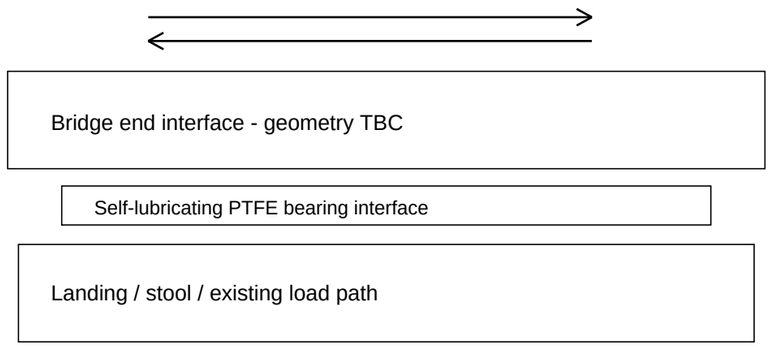
SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

ICW-ICG LD - BEARING AND LANDING INTERFACE DEVELOPMENT

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

FUNCTIONAL SUPPORT DIAGRAM - COMPONENTS NOT SIZED OR SELECTED

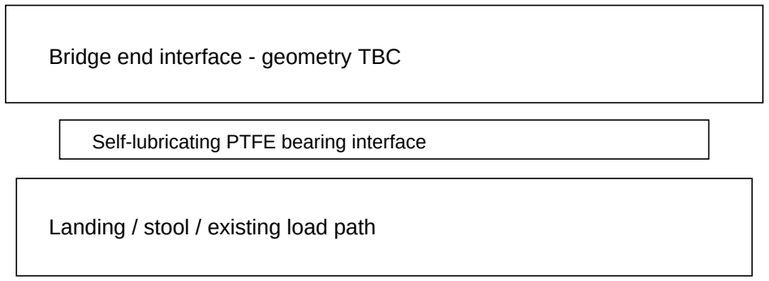
ICW PLATFORM | SLIDING



Longitudinal movement + lateral/vertical guides

ICG PLATFORM | HINGED / PINNED INTENT

Axial thrust and rotation provision to be designed



Does not imply moment-rigid frame-end restraint

MOVEMENT AND FORCE DESIGN BASIS

Required directional translation =1.5 x (absolute platform-A translation + absolute platform-B translation).

Required hinged-connection axial action =1.5 x expected axial thrust, with appropriate signed action envelope.

Thermal sensitivity deltaL =alpha x L x deltaT; compatible combined movement must avoid double counting.

Thermal sensitivity for this study length = 0.672 mm per degree C; actual design temperature range remains to be set.

Criteria location-tolerance wording must be interpreted for existing supports; it is not an adopted bearing-travel value.

DETAILS TO CLOSE BEFORE COMPONENT SIZING

Coordinates and datum; rotations; signed concurrent reactions; uplift and friction; travel and guide gaps.

Bearing pressure/contact; local plates and stiffeners; anchorage/connection resistance; corrosion and inspection access.

Surveyed existing steel / UT condition; landing/deck/jacket capacity; installation tolerances and replacement method.

Use correct platform drawing and verify member/interface labels before detailed modelling.

SHV-BRRP-101-BI-P4 | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

DC3.4 PartI3.4.16.4.17 pp42-43; source arrangement and platform drawings; dimensions pending

No member resistance, connection, bearing or lift approval is implied. Source results are not a new independently solved analysis.

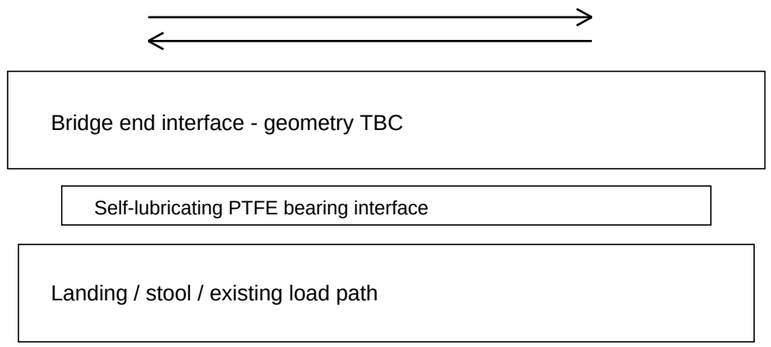
SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

ICP-ICG LD - BEARING AND LANDING INTERFACE DEVELOPMENT

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

FUNCTIONAL SUPPORT DIAGRAM - COMPONENTS NOT SIZED OR SELECTED

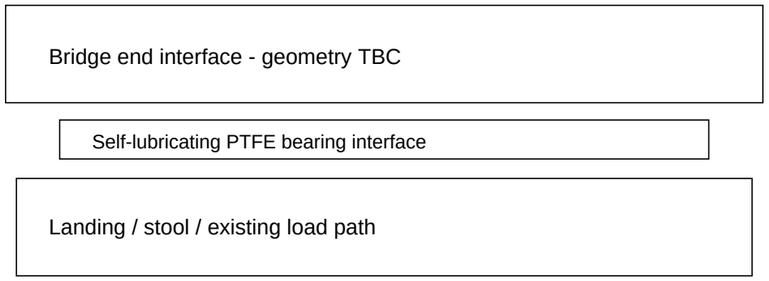
ICP PLATFORM | SLIDING



Longitudinal movement + lateral/vertical guides

ICG PLATFORM | HINGED / PINNED INTENT

Axial thrust and rotation provision to be designed



Does not imply moment-rigid frame-end restraint

MOVEMENT AND FORCE DESIGN BASIS

Required directional translation =1.5 x (absolute platform-A translation + absolute platform-B translation).

Required hinged-connection axial action =1.5 x expected axial thrust, with appropriate signed action envelope.

Thermal sensitivity deltaL =alpha x L x deltaT; compatible combined movement must avoid double counting.

Thermal sensitivity for this study length = 1.008 mm per degree C; actual design temperature range remains to be set.

Criteria location-tolerance wording must be interpreted for existing supports; it is not an adopted bearing-travel value.

DETAILS TO CLOSE BEFORE COMPONENT SIZING

Coordinates and datum; rotations; signed concurrent reactions; uplift and friction; travel and guide gaps.

Bearing pressure/contact; local plates and stiffeners; anchorage/connection resistance; corrosion and inspection access.

Surveyed existing steel / UT condition; landing/deck/jacket capacity; installation tolerances and replacement method.

Use correct platform drawing and verify member/interface labels before detailed modelling.

SHV-BRRP-102-BI-P4 | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

DC3.4 PartI3.4.16.4.17 pp42-43; source arrangement and platform drawings; dimensions pending

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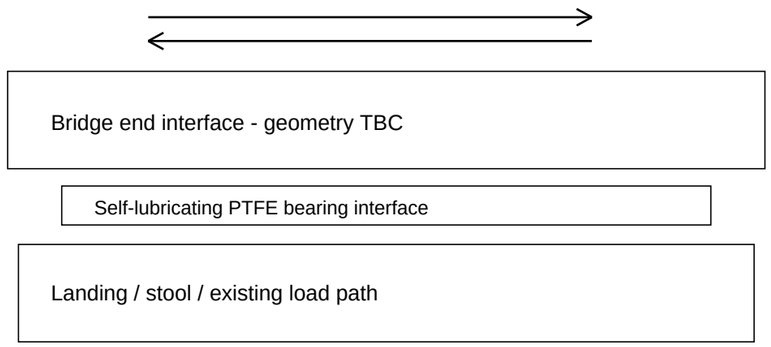
SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

BHS-SLQ LD - BEARING AND LANDING INTERFACE DEVELOPMENT

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

FUNCTIONAL SUPPORT DIAGRAM - COMPONENTS NOT SIZED OR SELECTED

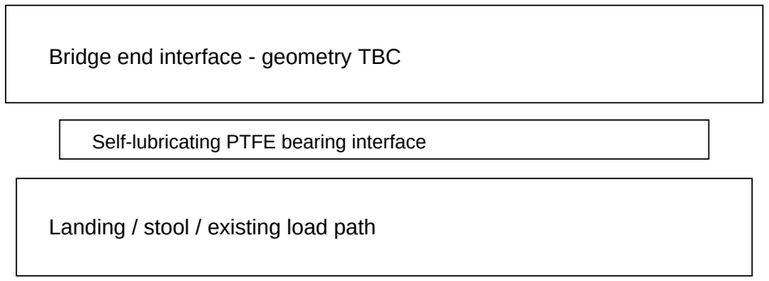
BHS PLATFORM | SLIDING



Longitudinal movement + lateral/vertical guides

SLQ PLATFORM | HINGED / PINNED INTENT

Axial thrust and rotation provision to be designed



Does not imply moment-rigid frame-end restraint

MOVEMENT AND FORCE DESIGN BASIS

Required directional translation =1.5 x (absolute platform-A translation + absolute platform-B translation).

Required hinged-connection axial action =1.5 x expected axial thrust, with appropriate signed action envelope.

Thermal sensitivity deltaL =alpha x L x deltaT; compatible combined movement must avoid double counting.

Thermal sensitivity for this study length = 0.504 mm per degree C; actual design temperature range remains to be set.

Criteria location-tolerance wording must be interpreted for existing supports; it is not an adopted bearing-travel value.

DETAILS TO CLOSE BEFORE COMPONENT SIZING

Coordinates and datum; rotations; signed concurrent reactions; uplift and friction; travel and guide gaps.

Bearing pressure/contact; local plates and stiffeners; anchorage/connection resistance; corrosion and inspection access.

Surveyed existing steel / UT condition; landing/deck/jacket capacity; installation tolerances and replacement method.

SLQ conflict remains a query: report/methodology fixed intent versus contradictory SOW wording.

SHV-BRRP-103-BI-P4 | Rev P4 | 25 Oct 2026 | Source-based development sheet; new Shivansh ID

DC3.4 PartI3.4.16.4.17 pp42-43; source arrangement and platform drawings; dimensions pending

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SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

ICW-ICG LD - DETAILING AND CALCULATION INTERFACE SCHEDULE

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

PROPOSED DETAIL MARKS - ASSIGN TO FINAL MODEL AND CALCULATIONS BEFORE DRAWING RELEASE

Mark	Detail family	Calculation / evidence required	Current status
CH	Chords	Section/group mapping; axial/bending/buckling/fatigue	NOT SELECTED
BR	Braces / crossframes	Node geometry; effective lengths; joint capacity	NOT SELECTED
WB	Walkway beams	Actual spans; local loading; grating support	NOT SELECTED
EC	End connection	Concurrent forces; rotations; local resistance	NOT SIZED
BE	Bearing / guides	Movement; pressure; uplift; friction; anchorage	NOT SIZED
LS	Landing / stool	Survey/UT; support reactions; existing capacity	VERIFY SOURCE
SS	Services supports	Routing; operational/hydrotest loads; future allowance	COORDINATE
AC	Access / fittings	Clear width; handrails; grating; standards/IRT	SOURCE DETAILS
TW	Temporary works	Stage loads; support arrangement; stability	NOT DESIGNED
LP	Lifting interface	Verified mass/COG; rigging; skew/dynamics	NOT DESIGNED
SF	Transport restraints	Vessel motions; support reactions; deck checks	NOT DESIGNED

INSPECTION AND DOCUMENT REQUIREMENTS

Material traceability and certificates; approved welding procedures / qualifications; dimensional control.

NDT extent, hold/witness points and acceptance per applicable specification and approved inspection plan.

Coating exposure/system and inspection; source MTO IRT pp15-34; no generic welding or NDT percentage assumed.

Every final dimension/detail must identify controlling source, calculation reference, revision and engineering reviewer.

BRIDGE-SPECIFIC INTERFACE

IC complex: landing drawing references, addendum jacket strengthening, access and adjacent live-service clearance.

SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

ICP-ICG LD - DETAILING AND CALCULATION INTERFACE SCHEDULE

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BRIDGE-SPECIFIC INTERFACE

IC complex: landing drawing references, addendum jacket strengthening, access and adjacent live-service clearance.

SHIVANSH / ONGC BRRP | PRELIMINARY DESIGN DEVELOPMENT

BHS-SLQ LD - DETAILING AND CALCULATION INTERFACE SCHEDULE

P4 - FOR ENGINEERING REVIEW | CONSTRUCTION RELEASE DEFERRED | NOT FOR PROCUREMENT OR EXECUTION

PROPOSED DETAIL MARKS - ASSIGN TO FINAL MODEL AND CALCULATIONS BEFORE DRAWING RELEASE

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Coating exposure/system and inspection; source MTO IRT pp15-34; no generic welding or NDT percentage assumed.

Every final dimension/detail must identify controlling source, calculation reference, revision and engineering reviewer.

BRIDGE-SPECIFIC INTERFACE

BHS: piping/cable supports, roofing, service transfer, lifeboat/muster interfaces and demolition package.